DERIVATIONS OF A NODAL CURVE ARE EXACTLY DERIVATIONS OF ITS NORMALIZATION FIXING THE POINTS OVER NODES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the split-node setup, the node exact sequence, the Leibniz/annihilator computation, the extension fixing x' and x", and the converse subring preservation are all retained verbatim.

Editorial note. The label lemma-nodal-vector-fields names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Split nodes over algebraically closed fields. Same chapter, Definition definition-split-node and Lemma lemma-split-node. Every node of X has two rational inverse images on the normalization since k is algebraically closed.
- The node exact sequence. Same chapter, Lemma lemma-multicross. Supplies 0 -> O_{X,x} -> O_{X^nu,x'} (+) O_{X^nu,x"} -> k -> 0, exhibiting the completed local ring at the node inside the product of the two branches.

Source reference. Original reference 0CBV: sources/stacks-curves/pinned-original-source.tex, source lines 4572–4572. Prerequisite pointer invoked at proof line(s) 6932; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1X: sources/stacks-curves/pinned-original-source.tex, source lines 3478–3478. Prerequisite pointer invoked at proof line(s) 6940; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CBW: sources/stacks-curves/pinned-original-source.tex, source lines 4585–4585. Prerequisite pointer invoked at proof line(s) 6933; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Lemma 0.1. *Let k be an algebraically closed field. Let X be an at-worst-nodal, proper, connected 1-dimensional scheme over k. Let $\nu : X^\nu \to X$ be the normalization. Let $S \subset X^\nu$ be the set of points where ν is not an isomorphism. Then*

$$Der_k(\mathcal{O}_X, \mathcal{O}_X) = \{D' \in Der_k(\mathcal{O}_{X^\nu}, \mathcal{O}_{X^\nu}) \mid D' \text{ fixes every } x^\nu \in S\}$$

Proof. Let $x \in X$ be a node. Let $x', x'' \in X^\nu$ be the inverse images of x. (Every node is a split node since k is algebriacally closed, see Definition 0CBV and Lemma 0CBW.) Let $u \in \mathcal{O}_{X^\nu, x'}$ and $v \in \mathcal{O}_{X^\nu, x''}$ be uniformizers. Observe that we have an exact sequence

$$0 \to \mathcal{O}_{X,x} \to \mathcal{O}_{X^\nu, x'} \times \mathcal{O}_{X^\nu, x''} \to k \to 0$$

This follows from Lemma 0C1X. Thus we can view u and v as elements of $\mathcal{O}_{X,x}$ with $uv = 0$.

Let $D \in \mathrm{Der}_k(\mathcal{O}_X, \mathcal{O}_X)$. Then $0 = D(uv) = vD(u) + uD(v)$. Since (u) is annihilator of v in $\mathcal{O}_{X,x}$ and vice versa, we see that $D(u) \in (u)$ and $D(v) \in (v)$. As $\mathcal{O}_{X^\nu, x'} = k + (u)$ we conclude that we can extend D to $\mathcal{O}_{X^\nu, x'}$ and moreover the extension fixes x'. This produces a D' in the right hand side of the equality. Conversely, given a D' fixing x' and x'' we find that D' preserves the subring $\mathcal{O}_{X,x} \subset \mathcal{O}_{X^\nu, x'} \times \mathcal{O}_{X^\nu, x''}$ and this is how we go from right to left in the equality. □